

Drift Assignment — Implementation Log

Chronological execution journal — what got built, when, and why. Reconstructed from git history at the tip of the assignment (16 commits across **2026-07-04** → **2026-07-05**). Screenshots omitted; the doc bundle carries the visual proof separately.

How to read this log: each phase entry lists the ship date, the goal, what concretely landed, one or two non-obvious decisions worth calling out to a reviewer, and how the phase was verified before moving on.

Phase 0 — Initial Project Setup

Date completed: 2026-07-04 **Commit:** `1bb4db6` — *chore: phase 0 project setup — Android target, docs, structure, asmdefs*

Goal

Clean Android-ready Unity 6 project scaffolding, Unity MCP wired up, and the docs bundle skeleton so every phase has a place to write its output.

What was added

- Build target switched to **Android** — min SDK 24, ARM64 only, IL2CPP, .NET Standard 2.1, landscape-only orientation lock.
- `Assets/_Project/` folder tree (Scripts, Prefabs, ScriptableObjects, Scenes, Materials, Textures, Audio, Fonts) plus `Assets/ThirdParty/` for vendor packs.
- **6 Assembly Definitions** with enforced dependency direction: `Core ← Input ← Vehicle ← UI / Camera / Character`, `Damage → Core`.
- URP asset (Mobile) hand-tuned — shadow distance 40 m, cascades 1, HDR off, MSAA 2× — before writing any gameplay.
- `.editorconfig` at repo root mirroring Unity's official C# style guide.
- Cinemachine 3.1.7, Memory Profiler, Android Logcat, Recorder installed.
- CoplayDev Unity MCP registered; ping response verified.
- Docs bundle stubbed: REQUIREMENTS, ARCHITECTURE, IMPLEMENTATION_LOG, OPTIMIZATION, PROJECT_SETUP, CREDITS, TOKEN_USAGE, `build-pdfs.ps1`.

Non-obvious decisions

- **Assembly Definitions from commit 1**, not "when the project gets big". Locks the dependency graph so future phases can't reverse it, and speeds up incremental compile from the very first script.
- **URP tuned aggressively at Phase 0** — resisted the URP 3D template's defaults (shadow distance 150 m, MSAA 4×) so no phase inherits a bad baseline.

Verification

Empty scene builds a working `.apk`; Unity MCP `editor_state` returned `ready_for_tools: true`; docs bundle renders locally.

Phase 1 — Environment & Asset Setup

Date completed: 2026-07-05 **Commit:** 5f0529d — *feat(phase 1): desert circuit env — terrain, sand layers, HDRI, 1090m drift loop*

Goal

Believable desert environment matching the "Arabic Drifting" reference — long drivable circuit, dune horizon, warm sky.

What was added

- **500 × 500 m terrain** with sculpted heightmap (Perlin-based dunes, road corridor kept flat via bounds mask).
- **Two sand terrain layers** (Aerial Beach + Aerial Sand PBR sets from Poly Haven, CC0).
- **Kloofendal 43D Clear Puresky HDRI** as skybox (Poly Haven, CC0).
- **EasyRoads3D closed-loop circuit**, 1090 m end-to-end.
- Sun (directional light) tuned for warm desert exposure.

Non-obvious decisions

- **Chose EasyRoads3D over hand-authored spline mesh** — the free tier is enough for a closed loop, saved several hours of tooling.
- **Kept GroundSand005 as a fallback layer** in ThirdParty/ even though the Aerial pair was chosen — no build cost (not referenced by materials in scene), useful reference for a designer swapping the look.

Verification

Camera fly-through runs at target framerate in Editor; visual match with reference screenshots.

Phase 2 — Car Physics

Date completed: 2026-07-05 **Commit:** db748f5 — *feat(phase 2): car physics — RMCAR26 drivable, RWD/AWD/FWD, chase cam*

Goal

Drivable sedan with sim-cade drift feel, matching the "Arabic Drifting" reference car handling.

What was added

- **RMCAR26 prefab** wired as a RigidBody + 4 × WheelCollider under WheelFL/FR/RL/RR transforms.
- Scripts in DriftAssignment.Vehicle :
 - CarController — input → wheels orchestration
 - Drivetrain — torque split by DrivetrainMode { Fwd, Rwd, Awd }
 - GearBox — 6-speed auto + manual, RPM-based shifts
 - HandBrake — rear brake torque + stiffness cut for controlled slides
 - SteeringAssist — speed-sensitive angle + optional Helper assist blend
 - WheelController — visual rotation + steer for wheel meshes

- `IInputProvider` interface with `KeyboardInputProvider` implementation
- `CarConfig` `ScriptableObject` (baseline vehicle stats).
- Cinemachine `ChaseCam` following the car.

Non-obvious decisions

- `IInputProvider` **from day one** — touch, keyboard, and future replay all implement the same interface. Kept `CarController` untouched when the Phase 4 touch HUD replaced keyboard on-device.
- **Physics tuning exposed on `CarConfig` SO**, not inline `[SerializeField]` on `CarController` — the same knobs are designer-editable in the Inspector AND runtime-editable through the settings panel in Phase 5.

Verification

Drives, brakes, steers with WASD in Editor. Handbrake initiates drift. Framerate stable during long play sessions.

Interim — Track hazards, reverse gear, audio scaffolding

Date completed: 2026-07-05 **Commit:** 732be94 — *feat: track hazards, reverse fix, audio scaffolding, CLAUDE.md*

What was added

- Track-side props (barriers, cones, signs) placed around the EasyRoads3D loop.
 - Reverse-gear transition logic in `GearBox` — auto-shift into R when the driver brakes from a stop; auto-shift out of R into 1st on throttle.
 - `SoundLibrary` `ScriptableObject` scaffolding.
 - `CLAUDE.md` at project root with architectural notes + memory pointers.
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Phase 7 (part 1) — Damage system: dents, cascade, health

Date completed: 2026-07-05 **Commit:** c2a8985 — *feat(phase 7-8): damage system — dents, cascade, smoke, sparks, health HUD*

Goal

Two-stage damage — vertex-level dents feed a per-part accumulator; heavier impacts detach body panels.

What was added

- `ImpactReceiver` on car root — routes `OnCollisionEnter` impulses to nearby deformable meshes.
- `DentableMesh` — runtime vertex displacement in a radius around each contact, original vertex positions cached at `Awake` for reset.
- `DamageCascade` — routes accumulated impulse into detachable-part triggers.
- `ImpactVfx` — smoke + spark particle spawners driven by impact tier.
- `CarHealth` component + HUD health readout.

Non-obvious decisions

- **Dent radius + falloff curve carefully tuned**, not a hard boolean. A soft radial displacement reads as "denting" instead of "punched a hole" regardless of contact geometry.
 - **MeshCollider deliberately NOT updated** as the visual mesh deforms. The collider stays a static convex hull; only the visual mesh mutates. On mobile the cost of a per-frame `MeshCollider.sharedMesh = null` rebuild would kill the frame budget for zero gameplay benefit.
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Phase 7 (part 2) — Runtime paint spoilage + welder sparks

Date completed: 2026-07-05 **Commit:** `dd58373` — *feat(damage): runtime paint spoilage, welder sparks, event-normal wiring*

What was added

- `PaintDamage` — runtime paint texture that "spoils" (scratches, dulls) where the car has been hit. Assigned via **one shared material clone** to all 34 car body renderers.
- Welder-tier sparks: yellow-orange particles with cone shape, stretched billboard rendering.
- Contact-normal routing so paint spoilage aligns with the impact direction.

Non-obvious decisions

- **One shared clone material for paint, not per-renderer clones.** Each paint damage texture is ~4 MB; 34 clones would have cost **~136 MB of texture memory**. The one-clone trick saves that entire budget for a single-hero-car camera-focused demo.
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Interim — 4-layer engine mixer

Date completed: 2026-07-05 **Commit:** `77a2d9f` — *feat(audio): 4-layer RPM engine mixer with auto-start + Rotary X8 pack*

What was added

- `CarAudio` — 4 concurrent audio sources with triangular RPM envelopes: idle, low-mid, mid-high, high. Cross-fades between layers as RPM sweeps.
- Auto-start engine on Play (`Volume` sweep from 0 to full over 1.5 s).
- Rotary X8 audio pack imported into ThirdParty.

Non-obvious decisions

- **4-source mixer with envelopes**, not `PlayOneShot` per RPM update. A per-blip approach would fire unbounded voices under heavy throttle changes; the mixer holds voice count at exactly 4.
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Interim — Drift audio + skid marks

Date completed: 2026-07-05 **Commit:** `e365b98` — *feat(drift): handbrake sound, tire screech, skid marks + tire smoke*

What was added

- Handbrake engage / disengage SFX.
 - Tire screech loop, volume driven by lateral slip.
 - Skid trail decals per wheel, lifetime cap 8 s (bounded runtime mesh count).
 - Tire smoke particles under drift, spawn threshold on slip.
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Interim — Sand dunes + brick walls + impact filter

Date completed: 2026-07-05 **Commit:** 238bc6c — *feat(env+audio): sand dunes, brick walls, impact-tier picker, ground-contact filter*

What was added

- **Sand dunes** — procedural Perlin heightmap layered on top of the base terrain (55 m primary noise + 12 m ripples, road corridor preserved via safety pad).
 - **Boundary walls** — 4 walls (520 × 8 × 2 m) around the circuit with the Mixed Brick Wall PBR set (Poly Haven, CC0).
 - **Impact-tier picker** — bank selection based on impact tier (low / medium / high) with `PickRandom` from a modular SO bank.
 - **Ground-contact filter** in `ImpactReceiver` — rejects contacts whose normal is within 40° of world-up (car belly / wheel bounce during drift). Spares paint + spark + audio spam every time the car goes sideways at low ride height.
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Phase 3 — Camera cycle

Date completed: 2026-07-05 **Commit:** d2083a6 — *feat(camera): phase 3 — Cinemachine 3-cam cycle + hold-to-look-back*

Goal

Multiple driving perspectives cycled by a HUD button.

What was added

- Three Cinemachine cameras — `ChaseCam`, `CinematicCam`, `BroadcastCam` — plus a hold-to-look-back `LookBackCam`.
- `CameraRig` — cycles priority on button press; fires `CameraChanged` event.

Non-obvious decisions

- **Cinemachine priority pattern**, not manual `LateUpdate` position math. Only the active cam's Update runs; inactive cams cost ~zero per frame.
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Phase 8 — Post-processing

Date completed: 2026-07-05 **Commit:** 8c306c8 — *feat(post-fx): phase 8 — desert post-process profile (mobile-safe)*

Goal

Sensory polish that sells the physics without eating the mobile GPU budget.

What was added

- URP Post-processing profile with ****Bloom + Tonemap + Color Adjustments**
 - Vignette + Film Grain**.

Non-obvious decisions

- **Motion Blur / DoF / Chromatic Aberration explicitly disabled.** All three are documented mobile-GPU killers — even at "low" quality Motion Blur alone can cost 2 ms on Mali-class GPUs.
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Phases 3+4+5 — Touch HUD + Settings panel

Date completed: 2026-07-05 **Commit:** 8ad71dd — *feat(ui): phase 3+4+5 — Cinemachine cams, touch HUD, settings panel*

Goal

Complete the on-screen driving surface — steering, pedals, gear shifts, camera cycle, plus the settings surface for runtime tuning.

What was added

- **Touch HUD** (Canvas — Screen Space Overlay, safe-area aware):
 - Steering arrow buttons (analog via drag distance)
 - Gas + brake pedal buttons (press-hold)
 - Handbrake button
 - Gear $\pm$ shift buttons (manual mode) + gear text
 - RPM tachometer, speed KM/H, gear-mode label (AUTO / MANUAL)
 - Camera cycle button with icon + label swap on cycle
 - Settings gear icon
- `TouchInputProvider` implements `IInputProvider` from HUD widgets.
- **Settings panel** — full 6-tab surface: Transmission · Suspension · Camber · Audio · Graphics · About.
- `SettingsController` — explicit-field wiring (no reflection). Two-way binding UI $\leftrightarrow$ `TuningState` SO. Snapshotted defaults for a `Reset` button.
- Graphics tab lives-update `QualitySettings.shadows`, `UniversalAdditionalCameraData.antiAliasing`, `renderPostProcessing`, etc.
- `TmpLinkClickHandler` — makes `<link>` tags in TMP text clickable (About tab, credit URLs).

Non-obvious decisions

- `SetText(fmt, arg)` **on speed / RPM readouts** avoids the `"{...}"` interpolation allocation on every hot-path event. Wired from day one rather than added later during the GC audit.
 - **Rotated `RectTransform` gotcha** — the handbrake button was mysteriously ignoring pointer input; the fix was that a Y-rotated Image is invisible to the Graphics Raycaster. Saved to memory so it doesn't bite a future developer.
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Hotfix — Manual-mode gear buttons

Date completed: 2026-07-05 **Commit:** `367c304` — *feat(hud+gearbox): manual-mode gear buttons enable/disable + shift-out-of-reverse*

What was added

- `GearButtonsController` — subscribes to `TuningState.Changed`. In AUTO the $\pm$ gear buttons dim to 0.35 alpha and stop receiving raycasts; in MANUAL they fade back in.
 - `GearBox.Update` scoped the "in Reverse, ignore shifts" early-return to automatic mode only. In manual mode, shift-up out of Reverse works.
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Phase 9 — Optimization pass

Date completed: 2026-07-05 **Commits:** `4b614d2` (pre-opt baseline), `fede200` (post-opt)

Goal

Bring a working game down to shippable APK size + mobile framerate on low-end hardware. Every decision measured, not guessed.

What was added

- `StressTestCapture` — `ProfilerRecorder`-based 60 s stat sampler that writes JSON to `Doc/stress-capture/`. Two captures (before / after) logged as the ground truth for the perf diff.
- 12-item optimization catalogue — see [OPTIMIZATION.md](#) for the full catalogue with rationale and delta per row.

Headline outcomes

- **APK 100 → 53.5 MB** (−46.5 %)
- **FPS 90.8 → 102.6 avg** (+13.0 %)
- **Main-thread ms 11.7 → 10.4** (−11.3 %)
- **Runs smoothly on Samsung Galaxy M31** (Mali-G72, Feb 2020)

Non-obvious decisions

- **Static-batching flag on 91 env renderers**, but honestly noted that the Stats-overlay Batches count goes UP after batching (Stats artifact), not down. The honest cost proxy — SetPass calls — dropped 8.8 % at p99, and that's the number the doc leads with.

- **Editor-only memory regression called out** — the +10 % `system_used_memory` after the pass is the Editor cache growing from reimports, NOT a real regression. On-device Player build inverts the sign (Vorbis + ASTC-6×6 both use LESS RAM than the baselines).

Verification

Two independent 60 s stress captures, same route, same car, same input pattern. Full JSON preserved. APK deployed to a Samsung Galaxy M31 and drives smoothly.

Phase 6 — Preset system (0–13)

Status: ➤ deferred

The numbered driving-style presets were absorbed into the Phase 5 settings panel — sliders + drivetrain dropdown + transmission toggle cover the whole tuning surface without a separate picker UI. Building a preset SO carousel on top would have been UI busy-work without adding a real gameplay knob. Called out honestly; not shipped.

Phase 10 — Character enter / exit

Status: ➤ skipped

Marked as a stretch goal in the spec. Skipped in favour of finishing the Phase 9 optimization + docs bundle to a polished state — the interview prompt weights physics feel + damage + code architecture + mobile performance, not a character rig.

Phase 11 — Documentation finalization

Date completed: 2026-07-05 **Status:** 🔄 in progress at time of writing

What was added

- All `Doc/*.md` files reviewed and updated: `OPTIMIZATION.md` consolidated with page-1 headline summary + page-2+ deep dive; `IMPLEMENTATION_LOG.md` (this file) backfilled from git history; `CREDITS.md` filled Fonts + Audio placeholders; `TOKEN_USAGE.md` phase status refreshed.
- **PDF pipeline** — `Doc/build-pdf-chrome.ps1` — zero-npm, zero-pandoc MD → PDF converter using PowerShell 7 `ConvertFrom-Markdown` + Chrome headless. Reruns are safe.
- All 7 docs exported to `Doc/pdf/`.
- `CLAUDE.md` at project root reflects the actual final structure.

Verification

Every doc reviewed; PDFs render correctly at A4 with no truncation or overflowing code blocks.